

MARCOS RODRIGO TALAVERA



Ph.D. Telecommunication Engineer and AI Researcher specializing in deep learning for computer vision, with expertise in face recognition, video understanding, and vision-language models. Full profile, projects and publications at marcosrodrigot.github.io

CONTACT

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SKILLS

AI & Deep Learning

- PyTorch
- TensorFlow
- HuggingFace
- OpenCV
- LangChain
- Ollama

ML & Data Science

- Scikit-learn
- NumPy / Pandas
- Weights & Biases

Programming

- Python
- C / C++
- Bash
- SQL
- HTML / CSS / JS
- LaTeX

Infrastructure & Tools

- Linux
- Docker
- Kubernetes
- AWS
- Git

Languages

- Spanish (Native)
- English (C1 Certified)
- French (Basic)

SUPPLEMENTARY EDUCATION

- How Diffusion Models Work (2025), DeepLearning.AI
- Generative AI with LLMs (2024), DeepLearning.AI, Coursera, AWS
- Deep Learning Specialization (2023), DeepLearning.AI, Coursera
- English as a Second Language (2015–16), UCLA
- Campus Party Tech. Event (2013), UAH
- Cognitive Development (2010–12), SAPIENTEC
- Mathematical Thinking (2006–09), Esc. Miguel de Guzmán

EDUCATION

- 2020 – 2025
Universidad Politécnica de Madrid, UPM
Ph.D. in Communication Technologies
Thesis: Automatic Sports Video Summarization with Identity-Aware Highlight Selection. Grade: Cum Laude.
- 2018 – 2020
Universidad Politécnica de Madrid, UPM
M.Sc. in Telecom. Engineering
Thesis: Automatic Highlight Detection in Martial Arts Tricking Videos. Grade: 10/10.
- 2013 – 2018
Universidad de Alcalá de Henares, UAH
B.Sc. in Telecom. Systems Eng.
Thesis: Acoustic Bird Classification Using MFCC Feature Extraction. Grade: 9/10.

PROFESSIONAL EXPERIENCE

- 2020 – present
E.T.S.I. Telecomunicación, UPM
Post-graduate Researcher
Designed deep-learning pipelines for face recognition (CNNs, ViTs), video highlight detection (2D/3D CNNs, RNNs, Transformers) and multimodal video summarization (CLIP). Led R&D with Nokia and Airbus. Taught across three degree programmes, supervised two Master's theses and built a 17-node Kubernetes GPU cluster.
- 2018
Urban Planet
Sports Instructor
Led group classes, ensured safety protocols and provided customer service in English.
- 2017 – 2018
Indra Sistemas S.A.
Junior Programmer
Developed VoIP scripts for air-traffic management. Automated tasks and configured network equipment under Scrum.
- 2016 – 2017
DEYDE Calidad de Datos S.L.
Data Coder & Trainee Programmer
Built expert systems for postal-address coding and maintained master databases.
- 2013 – 2018
Self-employed
Private Tutor
Taught Maths, Physics, Chemistry and English to students preparing for university entrance.

PUBLICATIONS

- M. Rodrigo, E. Zhong, C. R. del-Blanco, C. Cuevas, F. Jaureguizar, N. García. "ViMoCLIP: Augmenting static CLIP representations with video-motion cues for animal-action recognition." *IEEE/CVF CVPR Workshops (CV4Animals)*, 2025.
- M. Rodrigo, C. Cuevas, N. García. "Text-guided sports highlights: a CLIP-based framework for automatic video summarization." *IEEE Access*, 2025.
- M. Rodrigo, C. Cuevas, N. García. "Comprehensive comparison between vision transformers and CNNs for face recognition." *Sci. Reports* 14, 21392, 2024.
- M. Rodrigo, C. Cuevas, D. Berjón, N. García. "Automatic highlight detection in videos of martial arts tricking." *Multimed. Tools Appl.* 83, 17109–17133, 2024.
- M. Rodrigo, E. González-Sosa, C. Cuevas, N. García. "UPM-GTI-Face: Impact of distance and masks in face detection and recognition." *IEEE AVSS*, pp. 1–8, 2022.

AREAS OF EXPERTISE

- Deep Learning
- Computer Vision
- Multimodal Learning
- Face Recognition
- Video Summarization
- Action Recognition
- Vision Transformers
- CLIP / VLMs
- Object Detection
- Diffusion Models
- LLMs
- GPU Clusters
- Teaching